gpeyt1-23 · Flexible pavement SN

GEO PE · Long Calculation Problems I — Topic 5: Earth Structures, Ground Improvement & Pavements

Problem

An AASHTO 1993 flexible pavement design gives a required structural number SN = 4.6 for 8.2 million ESALs (R = 95%, S0 = 0.45, ΔPSI = 1.7, subgrade MR = 6,000 psi). Layers: 4.0 in. HMA (a1 = 0.44); crushed-stone base (a2 = 0.14, m2 = 1.0); 8.0 in. granular subbase (a3 = 0.11, m3 = 0.9).

Most nearly, the minimum base thickness D2 is:

A. 14.0 in.
B. 14.6 in.
C. 20.3 in.
D. 16.3 in.

Given equations / data

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SN = a1D1 + a2D2m2 + a3D3m3
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Worked solution

1. a1D1 = 0.44×4.0 = 1.76
2. a3D3m3 = 0.11×8.0×0.9 = 0.792
3. D2 = (4.6 − 1.76 − 0.792)/(0.14×1.0) = 14.6 in. (round up in practice)

Final answer

14.6 in.

Why the other values are wrong

- **14.0 in.** — ignored the subbase drainage coefficient m3
- **20.3 in.** — forgot the subbase contribution
- **16.3 in.** — applied m3 = 0.9 to the base instead

References (Jayce's library)

- FHWA NHI-05-037 Geotechnical Aspects of Pavements, Eq. 5.40 (SN = a1D1 + a2D2m2 + a3D3m3); NCEES PE Civil Reference Handbook v2.1 §3.19 Pavements

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=uGGy0dkZaX4, https://www.youtube.com/watch?v=wBgMPZkg_IE (no creator values copied).

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